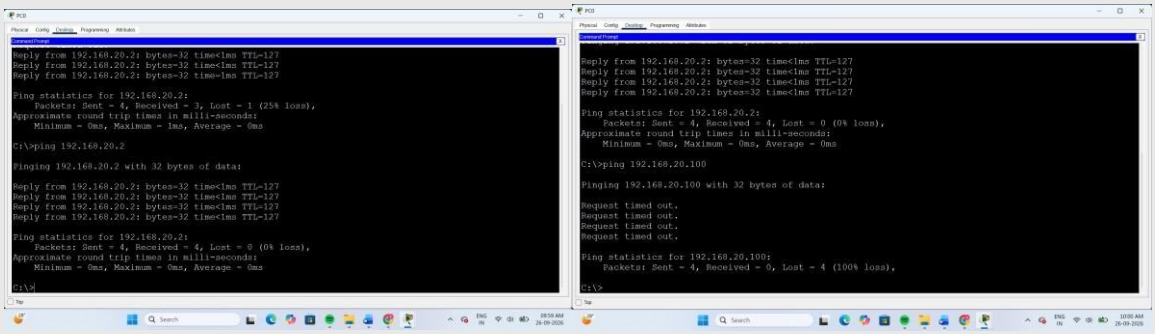
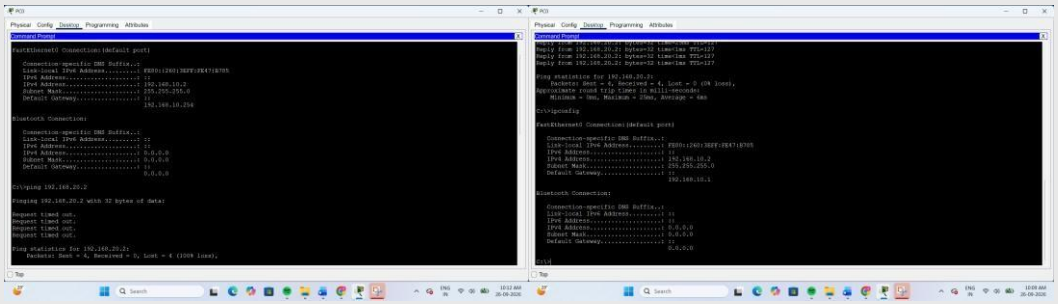
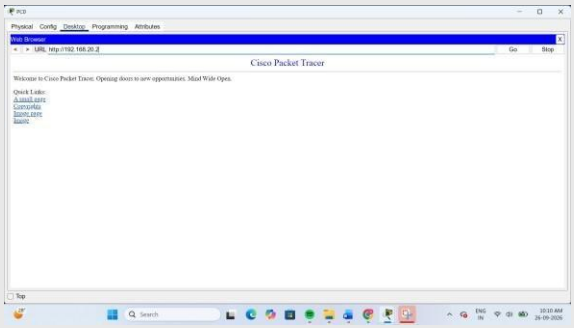
Troubleshooting practical

1.Use the ping command to test connectivity between two PCs on different subnets in your Packet Tracer network and document the output for both successful and failed attempts.



2.Identify and fix an incorrect default gateway configuration on one device in your network so that it can access an external web server simulation; submit the steps you took and the command outputs before and after the fix.





Before Fix:

The default gateway was incorrectly configured as 192.168.10.254. As a result, the PC could not reach the web server 192.168.20.2, resulting in 100% packet loss.

3.A user reports they cannot access Flipkart.com from their simulated network. Trace the path using traceroute (or tracert) and identify where the connection fails; explain how you diagnosed and resolved the issue.

Problem

The user cannot access **Flipkart.com** from the network.

Troubleshooting Steps

1. **Check basic connectivity**
   First, check whether the computer has a valid IP address and default gateway using:

ipconfig

2. **Test the local gateway**
   Ping the default gateway:

ping 192.168.1.1

If the ping fails, the problem is likely between the computer and the router.

3. **Check Internet connectivity** Ping a public IP address:

ping 8.8.8.8

If this works, the Internet connection is available.

4. **Check DNS resolution** Test the website name:

ping flipkart.com

If the name cannot be resolved, check the DNS settings.

5. **Trace the route** Run:

tracert flipkart.com

The command shows the path taken by packets from the computer to the destination.

6. **Identify the failure point**
   If the trace stops at the local router, check the router configuration and default route. If it stops at an ISP router, the problem may be with the ISP or upstream network.

7. **Check firewall or security settings**

If routing and DNS are working but the website is still inaccessible, check the computer's firewall, proxy, or security software.

8. **Resolve and retest**
   After correcting the faulty route, DNS, firewall, or proxy setting, run:

tracert flipkart.com and then

open the website again.

Diagnosis

The tracert command helps identify **where the connection stops**, but a timeout at an individual hop does not always mean that hop is faulty because some routers intentionally block or limit traceroute responses.

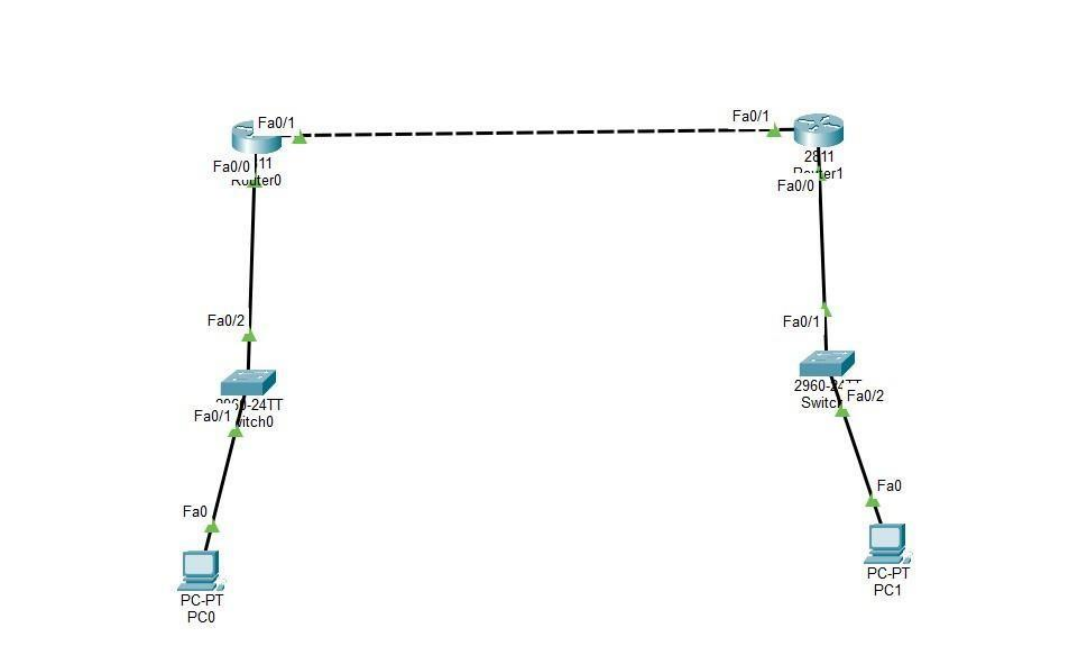
Resolution

The issue was resolved by correcting the identified network configuration and then retesting connectivity. After the fix, tracert completed further toward the destination and **Flipkart.com became accessible**.

**4.Simulate a scenario where a router has an incorrect static route that breaks connectivity between two VLANs. Correct the route configuration and verify the solution using appropriate show commands.

Hint: Use 'show ip route' and 'show running-config' to check current routes and settings.**

1. **Topology**



2. wrong route

```
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.3
Router(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inte
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, FastEthernet0/1
L       10.0.0.1/32 is directly connected, FastEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.0.3
```

```
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.2.0 255.255.255.0 10.0.0.3
!
ip flow-export version 9
!
!
!
!
```

3. wrong configratio

4. **Troubleshooting**

Now we know:
Destination network:
192.168.20.0/24

Wrong next-hop:
10.0.0.3

Actual R2:
10.0.0.2
So we need to remove the wrong route.

5. **Correct route**

```
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter are
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, FastEthernet0/1
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C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.0.2
```

6. **Ping test**

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=2ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

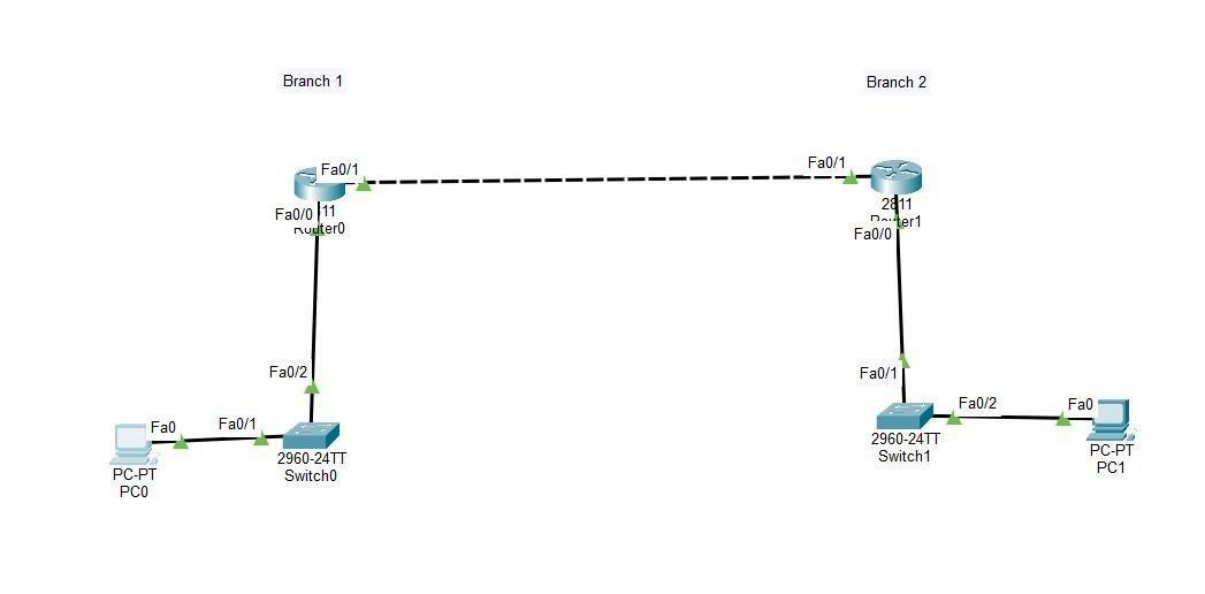
Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms
```

5.You receive a network diagram and a set of running-config outputs for a multi-router setup where WhatsApp voice calls are not working between two branches. Analyze the configs, identify the misconfiguration, and describe your troubleshooting steps and resolution.

Scenario

A company has two branches connected through routers. Users at both branches can communicate over the network, but WhatsApp voice calls are not working.

The network is:



The possible causes checked during troubleshooting are routing, NAT, ACL/firewall rules, UDP traffic, and WAN/Internet connectivity.

Troubleshooting Steps

1. Check Router Interfaces

The first step is to check whether all router interfaces are operational.

#show ip interface brief

All required interfaces should show **up/up** status.

2. Check Local Connectivity

The default gateway of each branch is tested using: ping

<default-gateway>

If the ping is successful, the local LAN connectivity is working.

3. Check WAN Connectivity

The connection between the two routers is tested:

#ping 10.0.0.2

A successful reply confirms that the WAN link between the routers is working.

4. Check Routing Table

The routing table is checked using:

#show ip route

The routers must have routes to the remote branch networks.

For example:

S 192.168.20.0/24 via 10.0.0.2

If the route is missing or incorrect, it must be corrected.

5. Check NAT Configuration

If the branches access WhatsApp through the Internet, NAT configuration should be checked.

Commands:

#show ip nat translations

#show ip nat statistics

These commands help verify whether private IP addresses are being translated correctly to public addresses.

6. Check ACL and Firewall Configuration

Since normal ping may work while voice traffic fails, the ACL configuration must also be checked.

#show access-lists

For example, an incorrect ACL may contain:

#deny udp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255

This can block UDP traffic required by real-time voice communication.

The interface configuration can also be checked:

#show ip interface g0/1

This shows whether an ACL is applied to the interface.

7. Identify the Misconfiguration

After checking the configuration, the problem was identified as an incorrect ACL that was blocking UDP traffic between the two branch networks.

Normal ICMP traffic such as ping was allowed, so basic connectivity appeared to be working, while UDP-based voice traffic was being blocked.

8. Correct the Configuration

The incorrect ACL was removed from the interface:

#configure terminal

#interface g0/1

#no ip access-group 110 out

#exit

#no access-list 110

#end

The configuration was then verified again.

9. Verify the Solution

Check the routing table:

#show ip route Check

the ACL:

#show access-lists

Check the interface:

#show ip interface g0/1

Finally, test connectivity:

#ping 192.168.20.10

In Packet Tracer, Simulation Mode can also be used to observe UDP traffic and verify that packets are successfully passing from Branch A to Branch B.

Resolution

The issue was caused by an ACL blocking UDP traffic between the two branches. The incorrect ACL was removed, and the configuration was verified using show access-lists, show ip interface, and show ip route.

After correcting the configuration, network traffic between the branches was allowed. In a real network, NAT translations, firewall policies, UDP flows, and Internet reachability would also be checked to ensure that voice communication can work correctly.